NeuroGuard AI - architecture as implemented (app.py, core/*.py, mcp_server/neuroguard_mcp_server.py)

All LLM roles: gpt-4o-mini, temperature 0, OpenAI-compatible Vocareum endpoint (OPENAI_API_BASE) | contract v1.0.0

Streamlit UI (app.py)
profile form, note, audience, memory consent | live trace, A2A log, MCP calls, checks clinician approve / reject report .md/.pdf + receipt.json

Input guardrails (core/guardrails.py)
PHI redaction -> prompt injection / tool misuse -> BE FAST emergency (call 9-1-1) -> dosing -> out of scope + profile validation | no LLM call

Run contract (harness_contract.json)
SHA-256 hashed before first token
30 LLM | 24 tool calls | 80k tokens
240 s | per-agent permissions
12 required checks

Orchestrator (LLM, core/agents.py)
structured output: next agent, skill, instruction, reason | code overrides: eligibility, FINISH only after review
caps: 9 turns, 2 visits/agent, 1 revision, budget; no tools

CaseState (LangGraph)
short-term memory per run: profile, intake, analyst, risk, evidence, retrieved, draft, review, turns, visits, trace[]

blocked -> refusal / emergency message (+ signed receipt, 0 LLM calls)

Task + skill

A2A-style message bus (core/a2a.py): Agent Cards, Task -> Message -> Artifact, inproc:// endpoints (no network)

orchestrator FINISH

Intake
skill: note-screening
MCP tools: screen_note, memory_read
(BE FAST signs, prior cases)

Risk Analyst
skill: cohort-comparison
MCP tools: predict_risk, explain_factors, cohort_stats, calculator | resource model://risk-card

Evidence Researcher
skill: guideline-citation
MCP tool: search_guidelines
+ resource guideline://{id}
reads of cited excerpts

Clinical Writer
skills: risk-communication | patient-friendly-summary
MCP prompt: clinical_report
no tools; retry once if finish_reason=content_filter

Safety Reviewer
skill: red-team-review
12-check invariant verifier (code)
+ LLM red team (prompt red_team_review), quote-confirmed | revise x1

Tool-using agents run the harness loop: context -> LLM (bind_tools) -> tool_calls -> pre-tool-use hook (allow/deny) -> MCP result (PHI-redacted) -> back to context, max 4 rounds

load_body()

prompts/get (allowed_prompt)

Permission engine + pre-tool-use hook
least privilege per agent (bind_tools + re-check); arg limits (k<=5, limit<=5, expr<=120 chars); tool budget; session_id pinned; memory_write needs consent; hash-chained audit log (core/permissions.py)

stdio JSON-RPC

MCP server: official mcp SDK (FastMCP), stdio subprocess
8 tools: predict_risk, explain_factors, cohort_stats, screen_note, search_guidelines, calculator (AST sandbox), memory_read, memory_write
4 resources: guideline://index, model://risk-card, dataset://stroke-card, policy://harness-contract
+ 1 resource template: guideline://{chunk_id}
2 prompts: clinical_report, red_team_review
client: core/mcp_client.py + langchain-mcp-adapters (sys.executable); OPENAI / NG_RECEIPT / GH keys stripped from server env

write

Finalize gate (graph node)
re-runs the 12 checks on the final draft; refuses to finalize if any fails
clinician sign-off if high risk, red flags or approved-with-warnings
memory_write via hook (consent)

Skills (skills/, core/skills.py)
6 SKILL.md: note-screening, cohort-comparison, guideline-citation, risk-communication, patient-friendly-summary, red-team-review | catalog = name + description only; full body injected when orchestrator selects it

Signed receipt (core/pipeline.py)
HMAC-SHA256 over contract, MCP server, inputs, report, trace hashes + audit head every run (incl. blocked); clinician decision re-signs it (prior sig chained)

P1 data + P2 statistics
Stroke Prediction Dataset (Kaggle, 5,109 rows) | core/data.py cleaning log
core/stats.py: chi-square, Welch t, odds ratios, Wilson CIs

P3 risk model (core/model.py)
LR / RF / GBM, class weights, 5-fold CV
served: calibrated LR (sigmoid)
exact log-odds explanations
threshold: OOF recall >= 0.80

P4 notes (core/notes.py)
CDC BE FAST rules (emergency)
TF-IDF + LR on MTSamples (CC0)
advisory neuro-note probability

P6 RAG (core/rag.py)
TF-IDF 1-2 gram + topic boost
30 CDC / NHLBI excerpts (public domain, with URLs)
data/guidelines.json

Long-term memory
SQLite (core/memory.py)
de-identified case summaries per session; consent + clear

P5 generative: Clinical Writer (LLM report from MCP prompt). Honest scope: A2A is in-process; data are public educational datasets; not a medical device.

Repo: https://github.com/sandipdey2/neuroguard-agentic-rag | Live demo: https://neuroguard-agentic-rag-4ywylkye6e2rgzavdb5xrb.streamlit.app/